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RESEARCH ARTICLE

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Bridging the theory-practice gap: a genealogical perspective on the most debated knowledge management challenges

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ABSTRACT

To address the theory-practice gap in knowledge management research, we identify the most persistent knowledge management challenges and proposed solutions over the past three and a half decades (1991–2023). Based on a systematic literature review combining a bibliometric and a qualitative content analysis, we conceptualise a genealogic (evolutionary) model to map the historical development path of three reappearing root challenges: (1) organisational antecedents to implement a knowledge-sharing culture (2) use of ICT for knowledge transfer and (3) knowledge conversion. We find that while topics 1 and 2 are established topics, topic 3 constantly renews itself by merging with innovation-related research streams. Further, our analysis reveals the general principle that the theory-practice gap opens at the end of the inception phase of a new challenge theme. We conclude that implementing artificial intelligence in knowledge management processes and fostering knowledge creation for sustainable development are the key topics for future research.

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Knowledge management challenges; genealogic perspective; theory-practice gap; historical development path; bibliometric analysis; qualitative content analysis

1. Introduction

An increasingly complex business environment gives rise to a knowledge-driven economy in which knowledge management (KM) is crucial for an organisation's success, competitiveness, and survival (Pauluzzo & Cagnina, 2017; Rix & Lièvre, 2008; Yeo & Marquardt, 2015). Knowledge management research discussed knowledge as a tool and a process, and KM as an organisational success factor or a social process (Gaviria-Marin et al., 2019). Just as continuously as the research is developing, practitioners point to KM implementation challenges, and over the last decades, scholars discussed many of them (Durst & Runar Edvardsson, 2012; Lee & Chen, 2012), for example, managing knowledge within virtual teams (e.g., Baker et al., 1997), organisational factors for successful KM implementation (e.g., Rodríguez-Gómez & Gairín, 2015), and the transfer of tacit knowledge (e.g., Baker et al., 1997; Blumentritt & Johnston, 1999).

However, this happens in a highly dispersed body of literature that still lacks a systematic consolidation and thus obscures the overall picture. The academic literature does not yet provide an overview of the most persistent challenges. It also does not provide evolutionary models for understanding the historical roots of these challenges, nor does it capture the forms in which they have appeared to date. As a result, it remains impossible to review and categorise the solutions proposed so far in the light of the underlying KM root challenges. This ultimately contributes to deepening the "theory-practice gap" that has been recognised in relation to the implementation of KM research (Syed et al., 2018), as it limits the ability of the researchers to identify solutions that worked in earlier periods to solve KM implementation challenges that were similar to those that practitioners face today. Thus, incorporating all available knowledge to advise practitioners and help resolve the challenges of implementing knowledge management is a struggle faced by researchers.

In this paper, we propose that a genealogical perspective (Ancelin-Bourguignon et al., 2020; Thompson & Kumar, 2021), i.e., an evolutionary account of the scholarly literature on KM challenges and solutions, can significantly contribute to establishing the lacking overview. Such research is essential as it can reduce the diagnosed theory-practice gap through increasing the legitimacy of academic counselling to practitioners by grounding academic advice in knowledge that has been accumulated over decades of discourse. We therefore present here research that identifies the academic literature over the last three and a half decades from 1991 to 2023 most debated

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KM implementation challenges and their solutions based on a systematic literature review. To establish a comprehensive evolutionary overview, we formulate a model that traces the historical development paths of the fundamental root challenges and theorise generic principles according to which the scholarly discussion developed. Thereafter, we conclude on research avenues. The remainder of this paper is structured as follows: Section 2 summarises the theoretical background for the research, while section 3 describes the research methodology. The findings are presented in section 4, discussed in section 5, and concluded on in section 6.

2. Theoretical background

Knowledge management can be defined as a set of practices that explain how to create, develop, retain, and apply knowledge to enable organisational learning and innovation (Syed et al., 2018). The development of KM as an academic field is well researched. Its emergence is traced back to the 1960s when the first authors linked knowledge to competitive firm advantage. Gaviria-Marin et al. (2019) summarise the development of knowledge management in four generations. While the first generation of KM research in the 1960s defined knowledge as a tool, the 1990s decade (second generation) understood it as a process (Gaviria-Marin et al., 2019). In the third generation, the 2000s decade, KM is highlighted as essential for organizations' success and in the fourth generation (2010's decade), KM is emphasised as a social process (Gaviria-Marin et al., 2019).

Two streams of literature specifically deal with KM implementation challenges faced by practitioners, as summarised in Table 1:

Two streams of literature specifically deal with KM implementation challenges faced by practitioners. Authors from the first stream review and classify general KM challenges to establish an overview and conceptualise challenge-type categories. They distinguish human or people-related challenges from those that concern systems or technology and others that emerge from organisational structures or operations (Durst & Zieba, 2019; Hislop et al., 2018; Syed et al., 2018). An approach developed by Remus (2012) differentiates KM challenges according to the implementation process phase when they appear (e.g., knowledge creation, sharing, transferring, or retention). A further classification describes the KM challenges as depending on the organisation type, studying their forms in private versus public sector organisations (Malik & Al-Toubi, 2018; Soakell-Ho & Myers, 2011) and in SMEs versus large organisations (Durst & Bruns, 2018). Within this research stream, KM challenges are described shortly in each category, thereby creating a generic overview of the literature (Kalkan, 2008).

The second stream is made up of a large body of management literature which studies very specific KM challenges embedded in three sub-streams related to organisational structures, systems, and people (Peltokorpi, 2017). Much research investigates structural organisational factors that facilitate or impede KM processes like knowledge creation and sharing (Margaryan et al., 2015; Rodríguez-Gómez & Gairín, 2015; Wolf et al., 2011; Yeo & Marquardt, 2015). Computer-mediated communication and information technology tools gave rise to a research sub stream on KM challenges related to specific technologies and systems (Margaryan et al., 2015; Rodríguez-Gómez & Gairín, 2015; Schwartz, 2007; Vuori et al., 2018). Studies on challenges concerning people in organisations focus on issues related to knowledge sharing in cross-cultural, interdisciplinary, and/or virtual environments (Maaninen-Olsson et al., 2008; Margaryan et al., 2015; Pauluzzo & Cagnina, 2017; Peltokorpi, 2017; Vuori et al., 2018).

Yet, based on the current research, it is difficult to understand which challenges are the most persistent ones and which are mainly management fads (Abrahamson, 1991): Although stream 1 seeks to consolidate and classify challenge types, it does not provide a historic evolutionary perspective. Such a perspective would however be needed to identify the historical roots of these challenges, and to capture the forms in which they have appeared so far. Further, the literature about specific challenges in stream 2 is

Table 1. Overview of literature streams dealing with KM challenges.

	Focus and Aim	Findings	Representative Authors
Stream 1: General KM Challenges	Identifying general KM challenge categories and creating a generic overview	Classifications of KM challenges • people, technology, or operations – related • KM process stage dependent • organisation type dependent	Syed et al. (2018); Durst and Zieba (2019) Remus (2012) Malik and Al-Toubi (2018); Durst and Bruns (2018); Ramos Cordeiro et al. (2023)
Stream 2: Specific KM Challenges	Investigate the nature of highly specific KM challenges	In-depth insights into KM challenges about • structural factors • information systems/technology factors • people factors facilitating or impeding KM processes in specific organisations	Yeo and Marquardt (2015) Rodríguez-Gómez and Gairín (2015); Vuori et al. (2018) Pauluzzo and Cagnina (2017); Peltokorpi (2017)

highly dispersed and lacks a systematic consolidation of these in-depth insights.

The current state of research delimits researchers' capacity to advise practitioners who face a KM implementation challenge based on an overview of solutions that worked in earlier periods to solve similar implementation challenges (Verma et al., 2020). This situation contributes to deepening the "theory-practice gap" that has been recognised in relation to the implementation of KM research (Booker et al., 2008; Ragab & Arisha, 2013; Serenko, 2013; Syed et al., 2018). Serenko (2021) highlight that KM research decreasingly includes practitioner contributions and shows a lack of field studies. Ragab and Arisha (2013) point out that KM research happens mainly on a conceptual level, making it hard for practitioners to use academic findings.

In this paper, we present a study that takes up the repeated calls for research to close the theory-practice gap (Kalkan, 2008; Ragab & Arisha, 2013; Verma et al., 2020). We propose that a genealogical perspective (Ancelin-Bourguignon et al., 2020; Thompson & Kumar, 2021) can significantly contribute to establishing the lack of overview and increase the legitimacy of academic counseling to practitioners by grounding academic advice in knowledge that has been accumulated over decades of discourse. A genealogical investigation is an evolutionary approach that traces the historical development of a research field and thereby allows identifying "the various root 'tributaries'" (Rasche & Chia, 2009, p. 714) of a discourse. Its value and contribution lie in the capacity of "explaining the form and origins of significant contemporary phenomena" (Maclean et al., 2016, p. 614). In our study, we use a systematic literature review over the past three and a half decades to consolidate extant literature into a genealogic perspective that traces the evolution of the most debated KM implementation challenges and related suggested solutions.

3. Methodology

A systematic literature review is a method that allows to identify, evaluate, and map the knowledge created about a research area (e.g., Gaviria-Marin et al., 2019; Singh et al., 2020). Identifying, analysing, and critically appraising earlier publications would allow us to arrive at the aspired genealogy and historical perspective on the emergence of KM challenges and solutions, and thus, the systematic literature review represented an appropriate method for our research (Snyder, 2019).

We followed the advice to use a maximum variation sampling strategy (Suri, 2013), which required identifying relevant publications and then "synthesize research findings in a systematic, transparent and reproducible way" (Snyder, 2019, p. 334; see also Tranfield et al., 2003). Our study incorporated two stages: First, we used a bibliometric analysis to collect and identify relevant papers. The bibliometric analysis explores the macropicture evolution path of a research field by, in addition to identifying relevant work, providing an overview of the connections among articles and the intersection of topics (in our case: KM challenges) without compromising on rigour or the number of relationships investigated (Singh et al., 2020; Van Eck & Waltman, 2017). However, bibliometric studies are not a substitute, but a complement to traditional methods of literature investigation in management research (Verma & Gustafsson, 2020; Wolf et al., 2022). Thus, in a second research phase aimed at an in-depth analysis of the content of the identified articles to answer our research question (Merli et al., 2018; Petzold et al., 2019; Snyder, 2019), we combined the bibliometric with a classical qualitative content analysis following the systematic literature review method (Tranfield et al., 2003). The research methodology is visualised in Figure 1.

The data for this study was extracted from the Scopus database. This selection was made based on earlier studies which found that the number of papers obtainable in bibliometric analysis from the Scopus database correlate to an extremely high degree with those available from the Web of Science ("R2 ≈.99"; Archambault et al., 2009, p. 1320). Still, Scopus was found to offer a slightly wider and more comprehensive coverage and offers a more detailed indexing than other databases such as Web of Science (Guz & Rushchitsky, 2009; Pham-Duc et al., 2023). Further, the Scopus database facilitates a bibliometric analysis with specific functions such as the journal name, type of document, year of publication, and h-index metrics for documents (Sweileh, 2018).

We searched for bibliographic records using the bibliographic fields article title, abstract and keywords. We used the following search string build of keywords proposed by Gaviria-Marin et al. (2019): ("knowledge management" or "organizational knowledge" or "knowledge acquisition" or "knowledge creation" or "knowledge integration" or "knowledge transfer" or "knowledge sharing or "knowledge diffusion" or "knowledge spill-over" or "knowledge use" or "knowledge application") AND ("challenge" or "risk" or "barrier"). Only peer-reviewed articles in English were retained, resulting in a final data set of 765 papers for the years 1991–2019.

Our study was designed to map the scientific literature on KM challenges of the last three decades. Even though KM research started to increase in publications after 1995 (see Figure 2), the first decade in this research starts with the year 1991 as this was the year when, according to the most recent review by Gaviria-Marin et al. (2019), one of the most influential articles

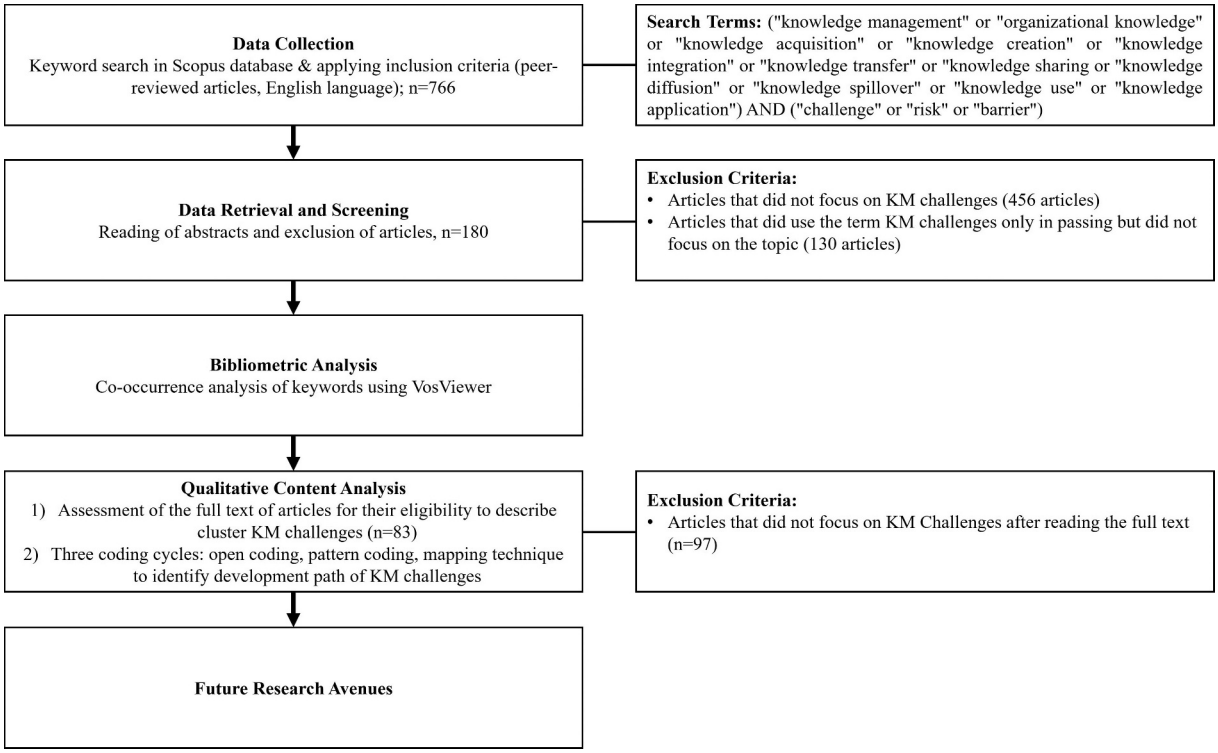


Figure 1. Overview of research design.

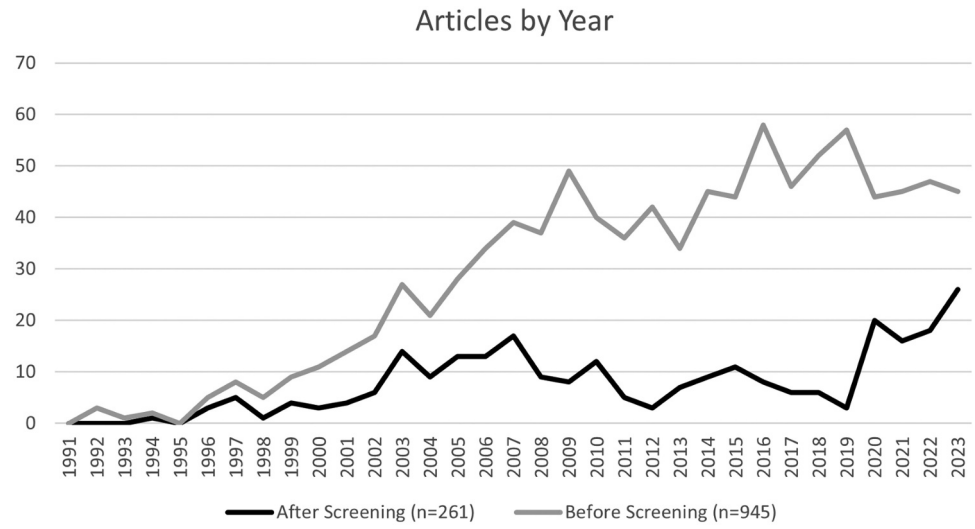


Figure 2. Distribution of KM challenges articles from 1991 to 2019.

in knowledge management research was published by Nonaka (1991). Further, the decades reflect Gaviria-Marin et al. (2019) decades of knowledge management development, as we similarly include the 1990s, 2000's, and 2010's decade. The bibliometric study identified the following numbers of articles per decade: Decade 1 that ranges from 1991 to 2000 includes 44 articles before/17 articles after the application of the exclusion criteria on the abstracts. Decade 2 (2001–2010) and Decade 3 (2011–2019) contain 306/105 and 414/58 articles, respectively. We used the bibliometric tool VOSviewer 1.6.8 and co-occurrence analysis to investigate semantic maps to understand the conceptual structure of challenges in KM. The graphical map is a network of items built through text mining, and the distance between items in a network is interpreted as indication of their relatedness. We arrived at four semantic clusters of KM challenges for decade 1, seven for decade 2, and six for decade 3.

In the second research phase, we subjected the datasets to a rigorous assessment with a qualitative content analysis, ensuring that the identified themes – in our case, KM challenge clusters – identified during the first research phase were in accordance with the main tenets presented in the literature (Callahan, 2014; Petticrew & Roberts, 2008). Based on reading the full text, we excluded further articles that did not focus on KM challenges, with a of total 83 articles

remaining–16 for decade 1, 40 for decade 2, and 27 for decade 3 (see Wolf et al., 2024). We adopted Krippendorff's (2013) content analysis methodology to ensure a robust analysis. To balance the risk of subjectivity in data analysis, we triangulated investigators by involving three researchers (Denzin, 1989; Flick, 2009). One researcher coded data segments that reported KM challenges in all articles, with an open coding approach to thematically classify text (Onwuegbuzie et al., 2016, p. 137). Two other researchers then reviewed these codes. After that, the three researchers came together and negotiated the meanings of first-level codes. In this step, no significant inconsistencies emerge. In the second coding cycle, the derived first-level codes were analysed for commonalities using pattern coding, a coding approach that allows for grouping codes by similarities (Onwuegbuzie et al., 2016, p. 138). Again, the researchers discussed the coding iteratively, and no significant inconsistencies arose. This step resulted in 12 pattern codes (Miles et al., 2014) representing the thematic clusters of KM challenges per decade – each decade consisting of four clusters.

In a final step, we compared the KM challenge clusters derived from the bibliometric analysis with those obtained from the qualitative content analysis. This step allowed for a comparison of research themes across the two analyses to further understand the evolution of the field and to uncover its contextual dimensions. Based on that, we merged and reorganised the 17 clusters from the bibliometric analysis into 12 clusters (four per decade) that mainly followed the structure identified with the qualitative content analysis. In a third coding cycle, we noted connections between the clusters across the decades using a mapping technique (Miles et al., 2014) to identify network relations embedded in the data (Saldaña, 2016) and used these insights to derive our genealogical development path model.

After analysing the last three decades, we conducted an additional literature review focusing on the first few years of the fourth decade (2020–2023) to update our research with the latest developments in knowledge management literature. The methodology for this review mirrored that of our earlier analysis. We again utilised the Scopus database, employing the same search strings. This initially yielded 181 research articles, which were narrowed down to 80 after applying the exclusion criteria (see Figure 1). VOSviewer 1.6.8 facilitated the creation of a semantic map, allowing us to visually interpret the relationships and clusters within the recent KM literature. The subsequent qualitative content analysis involved two researchers who assessed the themes and clusters identified in the bibliometric analysis by reviewing the abstracts, ensuring that the content was relevant to KM challenges. After reviewing the abstracts, additional articles were excluded from the study, resulting in a total of 38 articles included in this period. The content analysis followed the same process as for the last three decades, arriving at four pattern codes representing the thematic clusters of KM challenges in this period. In this analysis of the 2020–2023 period, which is shorter and not a complete decade, the focus is primarily on highlighting emerging trends and developments rather than offering a comprehensive overview like the previous decades. Consequently, this segment of the study provides a snapshot of the most current KM challenges. It enriches the overall research with up-to-date insights and future research areas but stands separately from the full-decade analyses.

4. Findings

The systematic literature review yielded sample-related descriptive findings and content-related clusters of the KM challenges and solutions.

4.1. Descriptive findings

Figure 2 presents the development of journal article publications on our topic over the last three decades identified within the bibliometric analysis. The grey line in the figure shows the number of articles in each year before screening for the article's relevance for this review, whereas the black line visualises the number of relevant articles in each year. For the screening process, the abstracts of all 945 articles were read and kept or discarded depending on the assessed relevance of the topic according to the research question, resulting in a total of 261 relevant articles. It can be identified that the peak of articles about KM challenges was in 2007, but that there were relevant publications throughout all years.

Table 2 provides an overview of the most influential journals publishing research on our topic.

While the Journal of Knowledge Management has published the most articles in the area, followed by Knowledge Management Research and Practice, the articles in the Strategic Management Journal generated the highest number of citations.

Table 2. Most influential journals publishing research on KM challenges.

Journal	Documents	Forward Citations
Journal of Knowledge Management	23	3446
Knowledge Management Research & Practice	9	250
Long Range Planning	4	964
Strategic Management Journal	3	5294
International Journal of Project Management	3	463
Journal of Product and Innovation Management	3	315

4.2. KM challenge clusters

4.2.1. Decade 1: 1991–2000

In the first decade from 1991 to 2000, four clusters were identified and verified during the qualitative content analysis (see Figure 3): 1) Strategically Managing Knowledge for Competitive Advantage (blue) 2) Implementing ICT for Distributed Knowledge Management (yellow) 3) Organizing Antecedents to Knowledge Management Implementation (red) and 4) Transferring Tacit versus Explicit Knowledge (green). The first cluster encompasses eight articles, the second cluster four, the third five, and the fourth six, while some articles were assigned to two clusters. With more than 70,700 citations, Nonaka (1991) can be regarded as the most influential researcher in this decade.

4.2.1.1. Cluster 1: strategically managing knowledge for competitive advantage. Several researchers pointed out the importance of strategic and efficient knowledge management as the key organisational capability to achieve competitive advantage (Blumentritt & Johnston, 1999; Dunford, 2000; Jordan & Jones, 1997; Quintas et al., 1997; Smith, 2000). With an increasingly global and volatile external environment, firms are forced to take on a more resource-based view. It is especially its knowledge and the capability to manage knowledge which can give the company a competitive advantage (Dunford, 2000; Jordan & Jones, 1997; Quintas et al., 1997).

The development leads to a more knowledge-based competition (Graham & Pizzo, 1996), and therefore, the main challenge for firms is to understand how they can manage knowledge strategically. Accordingly, articles in this cluster take on a more general and introductory perspective and describe what knowledge management comprises and which processes it includes (Blumentritt & Johnston, 1999; Jordan & Jones, 1997; Quintas et al., 1997). Jordan and Jones (1997) introduce the knowledge management assessment profile which is meant to serve as a starting point for managers to comprehend and manage their company's knowledge.

4.2.1.2. Cluster 2: implementing ICT for distributed knowledge management. This cluster deals with managing knowledge within globally dispersed teams and virtual companies. It focuses on the challenge of using information and communication technology (ICT) to support knowledge management in distributed and virtual environments (Baker et al., 1997; Goodman & Darr, 1998; Quintas et al., 1997). Dunford (2000) differs between input and output challenges of knowledge management systems that include ensuring good quality of information. They also point to the challenge that KM ICT tool implementation needs to be accompanied by a shift in mindset and culture "away from hoarding knowledge and towards sharing ideas" (p. 297). As one of the first empirical cases in this field, Baker et al. (1997) study of the internal consultancy of Royal Mail (UK) introduces a virtual tool and platform to communicate, share information, and solve problems across distributed departments. The researchers highlight that technology can be seen as critical to linking knowledge across a dispersed organisation, especially in those companies in which knowledge is the actual product, such as consulting.

4.2.1.3. Cluster 3: organizing antecedents to KM implementation. Scholars highlight that KM implementation depends on the presence of several supportive organisational factors (Heller, 2000; Skyrme &

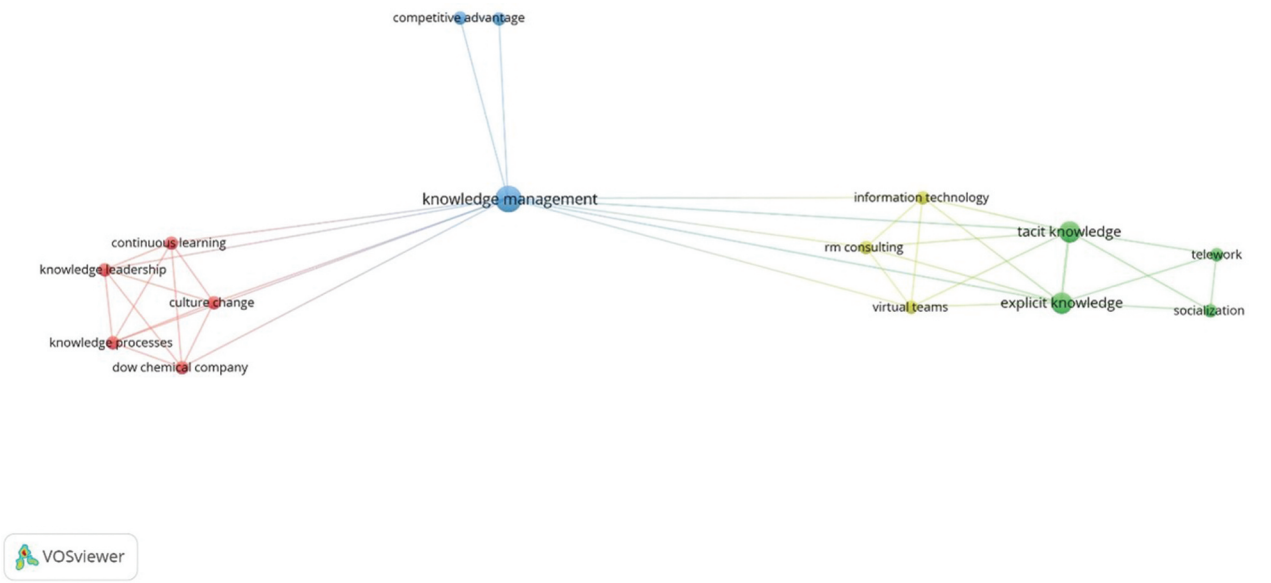


Figure 3. Visualization of the Co-Occurrence Analysis on KM challenges 1991–2000.

Amidon, 1997). Skyrme and Amidon (1997) point out that often challenges related to knowledge management are justifying the investment in KM, allow for gaining support from top management, and enable identifying KM leadership to carry out the knowledge agenda, creating a knowledge-sharing culture, fostering networks, and achieving the adaption of chosen supporting technology. Graham and Pizzo (1996) identify the challenge of balancing the organisational needs of the fluid (spontaneous, creative, experiential) and the more institutional (structures, codified, controlled, measured) side of KM.

Further, research in this cluster highlights that often, organisational culture needs to adapt or change to successfully implement KM (Graham & Pizzo, 1996; Heller, 2000; Skyrme & Amidon, 1997). Skyrme and Amidon (1997) suggest a diagnosis process to evaluate the knowledge agenda and the fit with the organisational culture. A strong link to business imperative, a compelling vision, individual champions, and existing KM initiatives to build on represent good prerequisites for pushing a knowledge agenda within a firm (Graham & Pizzo, 1996; Heller, 2000; Skyrme & Amidon, 1997).

4.2.1.4. Cluster 4: transferring tacit versus explicit knowledge.

Researchers in this cluster study the differences between tacit and explicit knowledge and investigate what challenges the two knowledge forms pose to knowledge transfer between employees (Baker et al., 1997; Blumentritt & Johnston, 1999). Raghuram (1996) compares the technical requirements for transferring tacit and explicit knowledge within telework. She finds that technological platforms enable the transfer of explicit but not of tacit knowledge in telework environments and highlights the importance of socialisation and mentoring for transferring tacit knowledge. Describing the transfer of explicit and tacit knowledge, Nonaka's (1991) influential article on the knowledge creating company introduces the knowledge spiral, which refers to creating knowledge through the conversion of tacit-to-tacit knowledge (socialisation), tacit to explicit (articulation), explicit to explicit (combination), and explicit to tacit (internalisation) knowledge.

4.2.1.5. Summary: KM implementation challenges, theories and solutions in decade 1.

Apart from KM implementation challenge descriptions, the literature also applies theories and outlines potential measures on how to deal with these challenges, as summarised in Table 3 below.

Like the KM challenges, the solutions presented in the scholarly debate are at a relatively generic level, requiring management attention and decision to kick-start a new culture and the implementation and use of ICT tools to deal with strategically relevant knowledge. The key theories closely associated with the challenges in the context of knowledge management include the resource-based view, technology implementation, leadership, organisational culture, and Nonaka's (1991) knowledge transfer spiral.

4.2.2. Decade 2: 2001–2010

The second decade (2001–2010) initially consisted of seven clusters that were identified through keyword co-occurrence analysis. From the cluster image (see Figure 4), we can identify a green cluster around

Table 3. Overview of cluster challenges, theories, and solutions for decade 1 (1991–2000).

Clusters: KM Challenges	Theories	Solutions	Representative Authors
1. Strategically Managing Knowledge for Competitive Advantage	Resource-Based View of Firms: This theory focuses on how firms can gain competitive advantage through the strategic management of their resources (in this case: knowledge).	• Understanding what knowledge management entails • Assessment of the company's knowledge management profile	Jordan and Jones (1997), Quintas et al. (1997), Blumentritt and Johnston (1999), Dunford (2000)
2. Implementing ICT for Distributed Knowledge Management	Factors for Successful Implementation of Information Technology: Identifies key elements necessary for effectively implementing IT solutions.	• Encouraging the use of technology by implementing a culture highlighting knowledge sharing instead of hoarding • Ensuring high quality of information • Actively managing knowledge database to avoid information overload	Baker et al. (1997), Dunford (2000)
3. Organizing Antecedents to KM Implementation	Supportive Organizational Factors for Knowledge Management Implementation: Highlights the role of leadership and organisational culture in facilitating knowledge management.	• Evaluation of the knowledge agenda and the fit with the organisational culture • Implementing a strong link to business imperative, a compelling vision, individual KM champions	Graham and Pizzo (1996), Skyrme and Amidon (1997)
4. Transferring Tacit vs Explicit Knowledge	Nonaka's Knowledge Spiral: Explores the process of converting tacit knowledge into explicit knowledge, emphasising the dynamics of knowledge creation and transfer.	• Technological platforms help transferring explicit knowledge • Socialisation and mentoring help transfer tacit knowledge	Nonaka (1991), Raghuram (1996), Baker et al. (1997), Blumentritt and Johnston (1999)

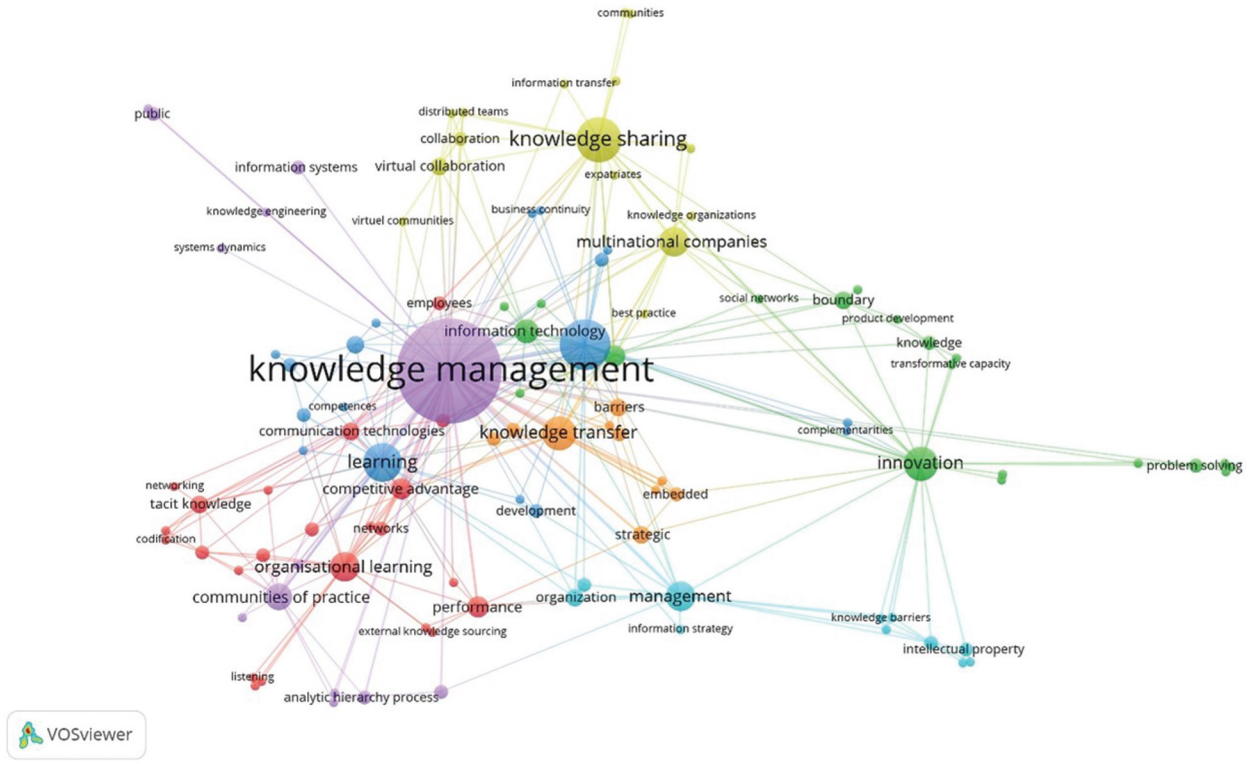


Figure 4. Visualization of the Co-occurrence analysis on KM challenges 2001–2010.

innovation and boundary spanning and a yellow cluster highlighting knowledge sharing in distributed and virtual teams. Further themes are knowledge barriers (teal cluster), knowledge transfer barriers (orange cluster), knowledge creation in communities of practice (purple cluster), organisational learning (blue cluster), and network management for knowledge sourcing (red cluster).

However, the content analysis of the respective articles resulted in merging three clusters. The remaining four distinct clusters are 1) Innovating across knowledge boundaries, 2) Creating an Infrastructure for Knowledge Transfer, 3) Organizing for a Learning Environment, and 4) Collaborating for knowledge creation in Communities of Practice. The first cluster includes seven articles, the second contains 14, the third comprises 10, and the fourth has nine articles. In this decade, the work by Carlile (2002) and Ardichvili et al. (2003) are the most cited ones, making the researchers the most influential ones in this decade.

4.2.2.1. Cluster 1: innovating across knowledge boundaries. Innovation is the central keyword of this cluster and is defined by scholars as a novel combination of knowledge which can be sourced internally or externally (Boder, 2006; Mors, 2010). Innovation is crucial for growth and competitive advantage and, at the same time, one of the main concerns of companies (Mors, 2010). Innovation and continuous learning can be facilitated by combining different knowledge bases (Maaninen-Olsson et al., 2008). Researchers in this cluster report KM challenges arising from the need to tap into various knowledge bases in innovation processes and specifically in new product development (NPD), to focus on multi-disciplinary teams and to work across functions (Carlile, 2002). In this context, the challenge is how to move knowledge across different knowledge boundaries. Carlile (2002) and Maaninen-Olsson et al. (2008) introduce boundary objects as a solution to this challenge: repositories, routines and rules, standardised forms, and methods, objects, or models. What connects NPD, problem-solving, and knowledge within this cluster is that NPD is seen as a dynamic problem-solving process (Marsh & Stock, 2006) that requires new knowledge to achieve its purpose (Boder, 2006).

4.2.2.2. Cluster 2: creating an infrastructure for knowledge transfer. Scholarly work in this cluster defines knowledge transfer as the process of effectively and efficiently moving knowledge within or between organisations (Bou-Llusar & Segarra-Ciprés, 2006; Schwartz, 2007). Communication is understood as the heart of knowledge transfer (Schwartz, 2007), and the level of difficulty in the transfer depends on the characteristics of the knowledge: The more tacit, complex, specific, and systemic the knowledge, the harder for the organisation to transfer it (Bou-Llusar & Segarra-Ciprés, 2006). However, as knowledge transfer is crucial in strategically managing knowledge, organisational barriers to it must be removed:

Schwartz (2007) categorises organisational barriers related to time, motivation, context, and design and barriers related to the recipient (absorption, retention) and the source (ease, ability, and willingness to communicate). Further, knowledge transfer is found increasingly difficult when deeply embedded routines and practices lead to rigidity and inhibit innovation processes (Collinson & Wilson, 2006). One proposed solution to the challenge of transferring knowledge is computer-mediated communication, which integrates otherwise standalone features like mails, chats, and forum systems (Schwartz, 2007).

To create an infrastructure for knowledge transfer, researchers underline the importance of a system perspective of KM which enables to explore different KM approaches in relation to each other (Garud & Kumaraswamy, 2005). The goal is to create a virtuous cycle of knowledge processes; therefore, the challenge is selecting and implementing KM through initiatives that fit the system (ibid; Anantatmula & Kanungo, 2010; Paroutis & Al Saleh, 2009). Lang (2001) points out that organisations should not collect and put knowledge on stock – instead, managers should focus on bringing people together, and on enabling and encouraging them to share knowledge. This requires a new IT and organisational architecture for managers to facilitate knowledge transfer. An IT architecture that includes "new languages, categories, and metaphors for identifying and accounting for skills and competencies oriented toward problem-solving and representation" (ibid, p. 56) and a new organisational structure described as "more social, transparent, open, flexible, and respectful of individual users" (ibid). IT and online forums are found as valuable tools to facilitate tacit knowledge transfer in multinational companies with more globally dispersed teams lacking face-to-face communication (DeSanctis et al., 2003; Majchrzak et al., 2005). Design principles for IT that can help to share content are studied by Majchrzak et al. (2005), whose key finding is that IT also needs to communicate context and thus support the following aspects of a contextualisation strategy: ownership, easy travel, multiple perspectives, indeterminacy, and the emergence of new categories.

4.2.2.3. Cluster 3: organizing for a learning environment. In this cluster, researchers focus on the challenges of considering and implementing an ideal workplace learning context and environment (Ley et al., 2008; Mehra & Dhawan, 2003). Organizational health, opportunities to learn, flexibility, risk-taking, innovativeness, and interaction are described as factors required to implement a thriving learning environment (Mehra & Dhawan, 2003). Ley et al. (2008) highlight work-integrated learning and point out the need to consider "knowledge workers' actual tasks, personal competency disposition and work domain as being relevant for deriving current learning needs" (p. 31). Existing knowledge can limit a knowledge worker's and organisation's learning capability (Andersen, 2008). In line with that, organisations find it hard to implement double-loop learning, as they most often only focus on single-loop learning or "refining patterns of behaviours" (ibid, p. 186). Organizations do not question core assumptions of behavioural patterns, and therefore, unlearning is a critical topic.

Learning plays a key role in project-based organisations, which creates challenges for knowledge to remain within the organisation, as every project is different and followed by a flow of personnel, materials, and information (Bresnen et al., 2003; Disterer, 2002). Factors to facilitate the retention of acquired knowledge and competencies, and consequently, the learning from projects include 1) reflection of activities at the end of a project (Disterer, 2002), 2) the supportive behaviour of leaders (Ayas & Zeniuk, 2001), 3) formal audits, memos, and presentations (Marsh & Stock, 2006), and 4) virtual communities to store acquired knowledge (Ordanini et al., 2008).

A few papers also deal with inter-organisational learning: Kidd et al. (2003) focus on learning in strategic alliances and Li and Tang's (2010) research outlines that "collaborations and partnerships can be a vehicle for new organizational learning" (p. 405).

4.2.2.4. Cluster 4: collaborating for knowledge creation in communities of practice. This cluster investigates collaboration for knowledge creation in virtual and local Communities of Practices (CoPs) (e.g., Garud & Kumaraswamy, 2005; Seba & Rowley, 2010). Chu et al. (2007) describe CoPs as the most effective organisational approach to cross-functional knowledge sharing and innovation as they "include managers and common knowledge workers who willingly and freely gather and agree their primary mission is to create, share, and apply knowledge to achieve a particular business goal" (p. 327). In addition to the organisational outcomes of CoPs, Ruuska and Vartiainen (2005) highlight personal (learning, new contacts) and group outcomes (trust, shared understanding, loyalty). Chu et al. (2007) introduce CoP assessment criteria, including locus of leadership, incentive mechanism, member interaction, and complementary assets. To implement effective CoPs, managers should identify potential informal communities and networks, help them form an official CoP, and provide the necessary organisational support (Hildrum, 2009). Support of CoPs includes measures related to their integration into the business, ICT support, and promotion and reward structure encouraging CoP participation (Hildrum, 2009). Ardichvili et al. (2003) study ways to remove

contribution barriers in virtual CoPs and conclude that knowledge-based and institution-based trust is required to facilitate participation. Virtual CoPs become especially important for multinational companies (MNCs). However, MNCs also need to consider the role of national culture for tailoring online communities (Ardichvili et al., 2006; Pan & Leidner, 2003).

In addition to CoPs, this cluster of scholarly work contains first contributions that highlight the benefits of a sound approach to network management for gaining access to external knowledge (Li & Tang, 2010). Network management that aims at benefiting all partners requires establishing firm routines for building and regularly refreshing meta-knowledge about the competencies and abilities that exist within the network (Evanschitzky et al., 2007).

4.2.2.5. Summary: KM implementation challenges, theories and solutions in decade 2. Table 4 provides an overview of the theories used and the solutions that researchers in Decade 2 propose to address the challenges in Knowledge Management (KM) implementation.

Other than in Decade 1 when the suggested solutions were very generic and mostly addressed to the top management, those proposed in Decade 2 are more concrete and involve also middle management. Still, when KM challenges related to leadership and/or organisational culture changes are discussed, the proposed solutions represent rather guidelines outlining which methods and approaches shall be used then specific procedures. Instead, data processing solutions for storing and transferring explicit knowledge are made at tool level.

4.2.3. Decade 3: 2011–2019

The co-occurrence analysis separated the KM challenges discussed in Decade 3 (2011–2019) into six clusters. The visualisation in Figure 5 shows that KM challenges in this decade relate to boundary spanning in networks (blue cluster), cultural barriers to knowledge sharing (purple cluster), knowledge creation (red cluster), learning (yellow cluster), and knowledge acquisition (teal cluster). Further, keywords cluster around knowledge managing, sharing, transfer, and application, initially describing the design ingredients of effective KM systems.

The content analysis of the articles resulted in the dissolving of three clusters (red, yellow, teal) and the creation of a new cluster. The four final clusters describe the KM challenges of 1) Spanning Boundaries in Largely Distributed Networks, 2) Overcoming Cultural Barriers to Knowledge Sharing for Innovation, 3) Organizing Factors to Knowledge Management Systems Implementation, and 4) Retaining Knowledge in Organizations. The first cluster comprises nine articles, followed by five in

Table 4. Overview of cluster challenges, theories, and solutions decade 2 (2001–2010).

Cluster: KM Challenges	Theories	Solutions	Representative Authors
1. Innovating across Knowledge Boundaries	Combining Different Knowledge Bases: Discusses how integrating various types of knowledge and using boundary objects can enhance understanding and problem-solving.	• Implementing boundary objects such as repositories, routines and rules, standardised forms and methods, objects or models	Carlile (2002), Maaninen-Olsson et al. (2008)
2. Creating an Infrastructure for Knowledge Transfer	Knowledge transfer: Identifies barriers and factors that facilitate the transfer of knowledge within organisations	• Computer-mediated communication integrating various features like mails, chats, and forum systems • Managers focusing on bringing people together, enabling and encouraging them to share knowledge within an IT and organisational architecture	Lang (2001), Garud and Kumaraswamy (2005), Bou-Llusar and Segarra-Ciprés (2006), Schwartz (2007), Paroutis and Al Saleh (2009)
3. Organizing for a Learning Environment	Learning Processes and Context/ Environment: Examines an ideal learning environment and context and how this influence the effectiveness of learning processes.	• Importance of organisational factors such as opportunities to learn, flexibility, risk-taking, innovativeness and interaction to implement a thriving learning environment • Project-related learning: reflection activities upon the ending of a project, supportive behaviour of leaders, formal audits, memos, and presentations	Ayas and Zeniuk (2001), Disterer (2002) Kidd et al. (2003), Mehra and Dhawan (2003), Marsh and Stock (2006), Ley et al. (2008), Ordanini et al. (2008)
4. Collaborating for Knowledge Creation in Communities of Practice	Effectiveness of Communities of Practice in Knowledge Sharing and Creation: Analyzes how communities of practice contribute to the sharing and creation of knowledge.	• Managers should identify potential informal communities and networks, help them form an official community of practice, and provide the necessary organisational support • Support of CoPs was identified through integration into the business, ICT support, and promotion and reward structure encouraging participation in physical and virtual CoPs	Ardichvili et al. (2003); Ardichvili et al. (2006); Chu et al. (2007); Garud and Kumaraswamy (2005); Seba and Rowley (2010) Hildrum (2009). Pan and Leidner (2003)

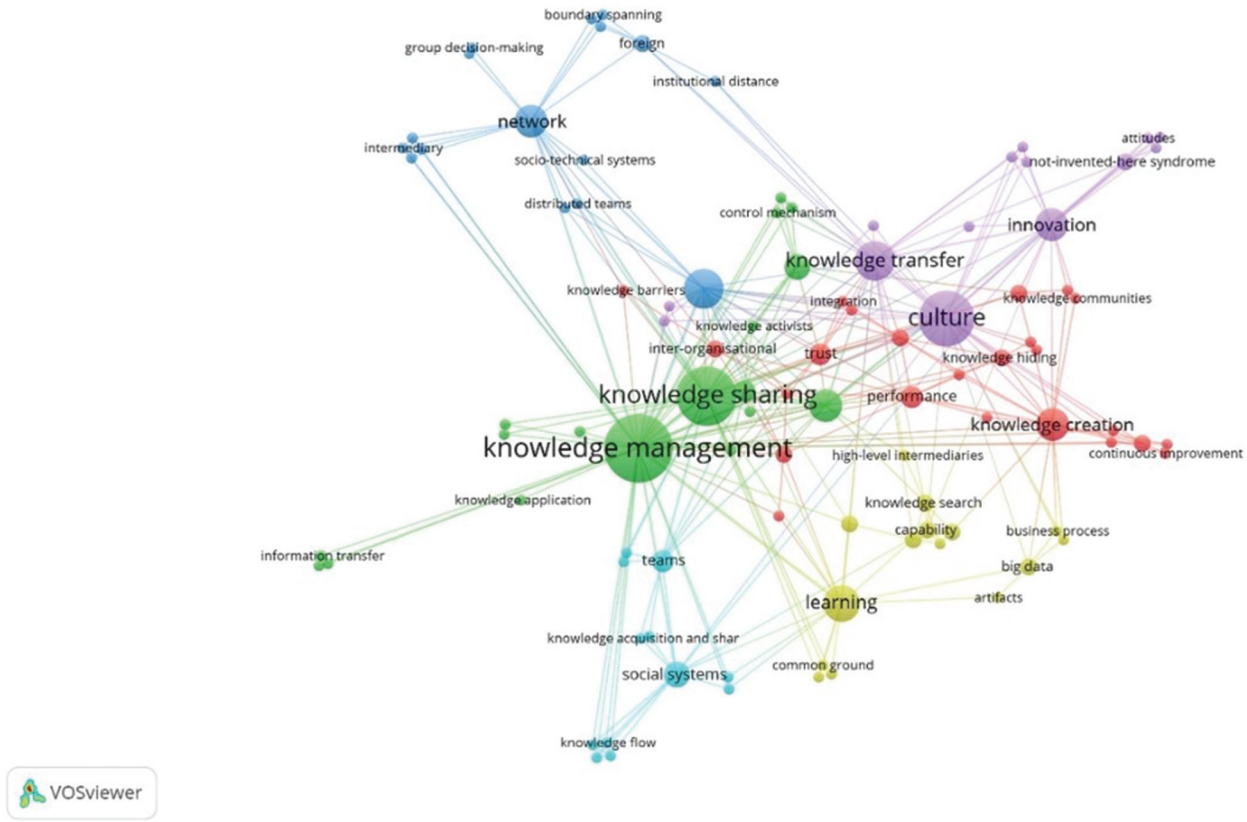


Figure 5. Visualization of the Co-Occurrence Analysis on KM challenges 2011–2019.

the second cluster, nine in the third, and four articles in the fourth cluster. With 425 citations, the work by de Araújo Burcharth et al. (2014) is the most cited. However, there are several other articles with similar citation counts (ranging from 200 to 300), including those by Levy (2011), Pandey and Dutta (2013), and Hwang et al. (2015).

4.2.3.1. Cluster 1: spanning boundaries in largely distributed networks. This cluster deals with knowledge boundaries emerging in inter- and intra-organisational networks. Consisting of functionally diverse and geographically dispersed members, networks are "made up of nodes and edges, where a node represents an entity, and an edge is a flow of some quantity between the nodes" (Broniatowski & Magee, 2017, p. 339). Organizations can act as innovation intermediaries that construct and manage extensive networks, allocate resources while making use of the network's database, and facilitate knowledge creation, storage, and transfer by promoting trust (Mount et al., 2015; Patriotta et al., 2013; Vuori et al., 2018).

Within an organisation, knowledge boundaries challenge geographically and temporally distributed teams as these need to achieve knowledge combination despite a lack of interaction and co-location of experts (Margaryan et al., 2015). In a case study, Margaryan et al. (2015) explain how the microblogging tool Yammer enabled distributed teams to successfully share "their current work, accomplishments, and issues" (p. 391) and create a feeling of connectedness. Within organisations and teams, status and personality differences in participants can create a challenge as individuals are more likely to seek knowledge from similar others but benefit most from knowledge held by those different from them (Haas & Cummings, 2015). Knowledge sharing in communities is found to be enhanced through experiential learning, because when individual learn about others' expertise, they interact more with those who are different than who are similar (Hwang et al., 2015)., Stephens and Carmeli (2016) investigate how discussions in complex teamwork can impact new knowledge creation. They highlight the importance of negative emotions being expressed constructively for new knowledge creation, and demonstrate that teams which share negative emotions perform better than others.

4.2.3.2. Cluster 2: overcoming cultural barriers to knowledge sharing for innovation. Articles in this cluster focus on barriers to knowledge sharing that arise from national and organisational culture and the not-invented-here (NIH) syndrome, which significantly poses challenges to innovation processes (de Araújo Burcharth et al., 2014; Grosse Kathoefer & Leker, 2012). The NIH syndrome is defined as preferring to use knowledge originating solely from inside

the unit or organisation, which can hinder knowledge transfer within or between organisations and impact knowledge creation (Grosse Kathoefer & Leker, 2012). In addition to the NIH syndrome, the not-shared-here syndrome negatively influences the use of open innovation practices in organisations (de Araújo Burcharth et al., 2014). Further, local subsidiaries might fear the undermining of their expertise by external sources of knowledge, implying that national culture can hinder the knowledge transfer in MNCs (Pauluzzo & Cagnina, 2017).

To overcome the NIH syndrome and the inherent challenges, de Araújo Burcharth et al. (2014) recommend innovation and creativity training as well as nursing special talents. Pauluzzo and Cagnina (2017) recommend solving the challenges emerging from national culture differences by developing intercultural competencies, especially in MNC's headquarters, to enhance trust and collaboration while strengthening the relationship with local subsidiaries. According to the authors, this can be facilitated by temporary teams, socialisation initiatives, training activities, and knowledge-sharing systems, thereby focusing on learning from other cultures and local subsidiaries. In addition, the willingness for knowledge sharing and transfer in MNCs can be fostered through a formal control mechanism and evaluation system that focuses on creating recognition and giving incentives for subsidiaries' knowledge sharing (Blomkvist, 2012).

4.2.3.3. Cluster 3: organizing factors for KM systems implementation.

Several researchers in this cluster investigate how and which organisational factors can facilitate the implementation of social KM ICT-supported systems such as wikis. The latter can create a knowledge flow by allowing "users to craft and recraft content, to build on others' contributions, to try out new ideas, to make knowledge visible, and to preserve content in ways that were difficult, if not impossible, in prior knowledge management systems" (Argyris & Ransbotham, 2016, p. 226). Wikis or IT systems promise simplicity, centrality, social effects, and communication to KM (Haamann & Basten, 2019). Challenges related to wiki implementation include a lack of user motivation and organisational tension (Argyris & Ransbotham, 2016). The most-mentioned facilitating organisational factors to support broad usage of wikis or IT systems are fitting leadership (e.g., Millar et al., 2016; Rodríguez-Gómez & Gairín, 2015), culture, and strategy (e.g., Sumbal et al., 2019). Argyris and Ransbotham (2016) introduce knowledge entrepreneurship as a new archetype of project leadership integrating managerial skills, technology affordances, and critical factors in KM processes. Pandey and Dutta (2013) point out that an organisational knowledge infrastructure capability is required for making KM effective – including cultural (mission, values), structural (collaborations across teams, rewards, and incentives), and technological capabilities.

4.2.3.4. Cluster 4: retaining knowledge in organizations.

Knowledge retention is described as a "unique sub-discipline of knowledge management" (Levy, 2011, p. 582) and deals with the challenges emerging from the loss of critical knowledge when employees leave the firm or when projects finish (Chandra et al., 2015; Mishra & Bhaskar, 2011). As employees' knowledge and expertise become organisational assets, organisations need to prepare for knowledge retention in three primary stages: 1) defining scope, 2) documenting, and 3) integrating knowledge back into the organisation (Levy, 2011).

Mishra and Bhaskar (2011) agree on the importance of a structured knowledge retention process and highlight the role of detailed exit interviews before employees leave. An essential aspect of knowledge retention is trying to capture the tacit knowledge of employees and experts to make it explicit to others (Chandra et al., 2015; Mishra & Bhaskar, 2011). Shipilov et al. (2017) investigate the possibility of organisations to benefit from building up a social or employee mobility network of key employees leaving the company. Their findings show that when employees leave the organisation to work for a foreign and lower-status competitor, organisations gain access to external knowledge benefitting the company while also facing a risk of internal knowledge leakage.

4.2.3.5. Summary: KM implementation challenges, theories and solutions in decade 3.

Table 5 summarises the theories and solutions that the studies in Decade 3 identified to approach the KM challenges debated in the research in that period.

In Decade 3, research proposes concrete solutions such as wikis or incentive systems to long-standing challenges such as knowledge-sharing motivation across boundaries. These are addressed to middle- and project managers. At the same time, new topics emerge such as knowledge retention or big data integration, and for those challenges, the proposed solutions resemble frameworks that can be used by the top management to initiate solution implementation.

4.2.4. Preliminary findings on decade 4: 2020–2023

The analysis of the articles published within 2020 and 2023 resulted in eight clusters (see Figure 6), and are centred around 1) knowledge transfer and project management (red), 2) information technology and collaboration (green), 3) risk management (dark blue), 4) artificial intelligence and innovation (yellow), 5) decision-making and knowledge application (purple), 6) sustainability and stakeholder engagement

Table 5. Overview of cluster challenges, theories, and solutions decade 3 (2011–2019).

Cluster: KM Challenges	Theories	Solutions	Representative Authors
1. Spanning Boundaries in Largely Distributed Networks	External and Internal Knowledge Boundaries: Focuses on managing networks involving multiple actors and sharing knowledge across geographically and temporally distributed teams	• Online KM tools • Familiarity through having a history of working together to encourage virtual knowledge exchange	Haas and Cummings (2015), Hwang et al. (2015), Margaryan et al. (2015)
2. Overcoming Cultural Barriers to Knowledge Sharing for Innovation	Not Invented Here Syndrome & Open Innovation: Addresses overcoming cultural barriers in innovation and enhancing intercultural competencies.	• Development of intercultural competencies to enhance trust and collaboration while strengthening the relationship with local subsidiaries: temporary teams, socialisation initiatives, training activities, and knowledge-sharing systems, • Formal control mechanism and evaluation system that focuses on creating recognition and thereby giving incentives for subsidiaries' knowledge sharing	de Araújo Burcharth et al. (2014), Blomkvist (2012), Pauluzzo and Cagnina (2017)
3. Organizing Factors to KM Systems Implementation	Importance of Factors in ICT Implementation and Usage: Stresses the significance of user motivation, leadership, culture, and strategy in the implementation and use of information and communication technologies.	• Facilitating organisational factors mentioned most throughout the articles in this cluster are fitting leadership, culture, and strategy • Knowledge infrastructure capability including a cultural (mission, values), structural (collaborations across teams, rewards and incentives), and technological capability	Pandey and Dutta (2013), Rodríguez-Gómez and Gairín (2015), Argyris and Ransbotham (2016), Millar et al. (2016), Sumbal et al. (2019)
4. Retaining Knowledge in Organizations	Knowledge Retention Processes: Explores methods and strategies for retaining critical organisational knowledge, particularly tacit knowledge.	• Preparation for knowledge retention by defining scope, documenting (planning and implementation), and integrating knowledge back into the organisation • Social or employee mobility network upon key employees leaving the company	Levy (2011), Mishra and Bhaskar (2011), Chandra et al. (2015), Shipilov et al. (2017)

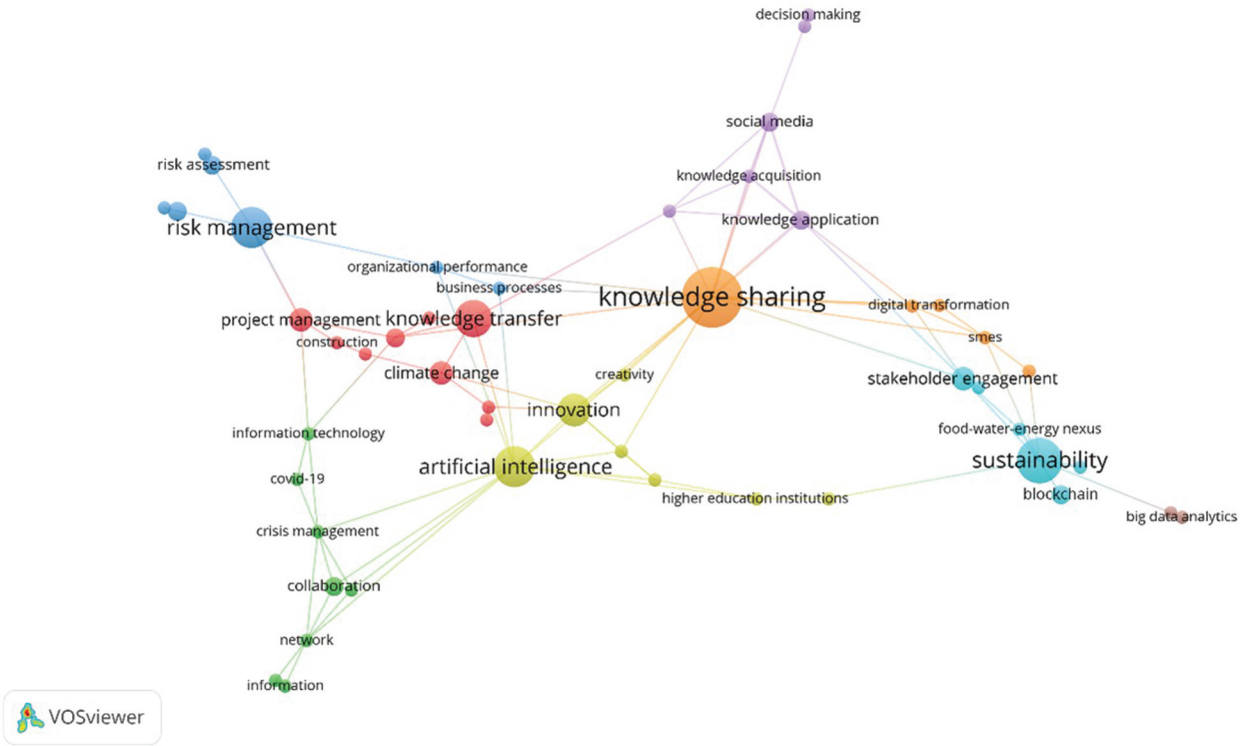


Figure 6. Visualization of the Co-Occurrence Analysis on KM challenges 2020–2023.

(light blue), 7) knowledge sharing (orange), and 8) big data analytics.

The content analysis of the papers resulted in merging several clusters so that in the end the following four clusters remain 1) Strategically Managing Knowledge to Mitigate Crises and Risks, 2) Navigating the Integration and Application of AI-Assisted Knowledge Management, 3) Leveraging Social Media and Digital Transformation for Knowledge Sharing, and 4) Creating Knowledge for Sustainable Development. The first three clusters each comprise six articles, while the fourth cluster contains nine articles.

4.2.4.1. Cluster 1: strategically managing knowledge to mitigate crises and risks. This cluster examines the strategic management of knowledge as a means to mitigate crises such as those caused by climate change and COVID-19 (Wang & Wu, 2021), as well as to navigate the complexities of internal risks, exemplified by the use of mobile shadow IT (Abbas and Alghail (2023). Several articles stress the necessity of crafting resilient knowledge strategies that are responsive to environmental and technological shiftsfor instance, regarding AI's potential role within healthcare supply chains (Ayakwah & Damoah, 2022) or how data analytics can increase construction safety (Xu & Zou, 2021).

The studies show that internal knowledge risks need to be discussed, as exemplified in the article by Abbas and Alghail (2023) on mobile shadow IT in organisations. This article presents a study that investigates the phenomenon of mobile shadow IT. It shows that when employees use unauthorised internet-based software and communication tools, this poses a threat to organisational knowledge protection due to potential security breaches and risks to data privacy. These studies highlight the need for organisations to refine their knowledge strategies, developing robust, adaptive, and secure systems to effectively respond to the rapid pace of change in the modern world.

4.2.4.2. Cluster 2: navigating the integration and application of AI-Assisted knowledge management. The articles in this cluster revolve around the integration of technology, particularly AI, in various domains, emphasising a transformation of knowledge management. The studies address the use of advanced technologies such as AI and machine learning to facilitate knowledge sharing, creation, and transfer (Ali et al., 2023; Nylund et al., 2023) through chatbot communication (Moldt et al., 2023). Common challenges identified include the complexities of effectively integrating AI to boost organisational performance (Ali et al., 2023; Olan et al., 2022), the imperative of fostering trust and addressing ethical dilemmas, and the critical importance of data privacy and secure interactions with AI systems (Moldt et al., 2023; Nylund et al., 2023).

A recurrent theme is the need for continued education and skill development to keep pace with AI advancements (Moldt et al., 2023). Moreover, the studies delve into the human dimensions of AI integration, considering AI's implications on human roles, pointing to a transformative impact on traditional responsibilities and the emergence of new behavioural dynamics in human-AI collaboration (Ali et al., 2023; Nylund et al., 2023; Olan et al., 2022).

4.2.4.3. Cluster 3: leveraging social media and digital transformation for knowledge sharing. This cluster of knowledge management research investigates the dynamics of knowledge sharing with a particular emphasis on the role of social media and digital transformation. For example, Panahi et al. (2021) and Gardeazabal et al. (2023) study the ways in which digital platforms encompassing social media and specialised knowledge management systems contribute to knowledge acquisition, sharing, and application. The articles recognise the challenges associated with the adoption of new technologies aimed at promoting knowledge sharing such as the need to adapt team structures, surmount knowledge-related obstacles, and ensure effective communication within technology-driven environments (Igwe & Olatokun, 2023; Zhang et al., 2022). Several authors address concerns regarding information reliability and privacy within the realm of social media (Panahi et al., 2021) or healthcare knowledge sharing (Igwe & Olatokun, 2023). Here, the accuracy and ethical use of shared knowledge emerge as critical issues. Moreover, this cluster explores the broader impact of digital technologies on knowledge management processes across diverse contexts, with a specific focus on the challenges and opportunities they introduce concerning knowledge sharing (Igwe & Olatokun, 2023; Panahi et al., 2021).

4.2.4.4. Cluster 4: creating knowledge for sustainable development. This cluster studies the challenges and strategies of integrating knowledge on sustainable practices and data analytics into various sectors, with a focus on stakeholders. A key focus is the integration of advanced technologies, such as blockchain and data analytics, to enhance sustainability practices (Frozza et al., 2023; Ratna et al., 2023). Frozza et al. (2023) present a conceptual model that positions blockchain as a critical factor in shaping the relationship between KM and sustainability, showcasing how it influences KM processes and ultimately contributes to organisational sustainability across multiple dimensions.

The articles underscore the role of knowledge management in achieving sustainability goals in institutions such as, for example, Nepalese higher

education institutions (Adhikari & Shrestha, 2023). The findings emphasise the need for a cultural shift, multidisciplinary approaches, improved regulations, and the incorporation of KM practices into academic programs to enhance higher education's contribution to SDGs. Overall, these articles point to the essential role of knowledge management in addressing the complexities of sustainability challenges and demonstrate the importance of engaging stakeholders and communities in sustainable development initiatives, thereby highlighting the significance of participatory methods for inclusive decision-making processes (Patra et al., 2021; Venegas et al., 2021).

5. Analysis and discussion

Earlier research has highlighted the importance to overcome the lack of practice-oriented KM research, i.e., research that takes up and addresses the challenges faced by practitioners (Kalkan, 2008; Booker et al., 2008; Serenko, 2021; Ragab & Arisha, 2013; Se; Syed et al., 2018; Verma et al., 2020). Particularly, a "lack of problem-driven KM research (. . .) suggest(s) that much of the KMRP the research work currently reported in may be quite far (too theoretical, too generic) from the actual work of KM practitioners" (Edwards & Lönnqvist, 2023, p. 912; see also Dumay, 2022; Jevnaker & Olaisen, 2022).

Taking up these calls for practice-oriented research, the in this paper conducted study aimed at contributing to reducing the diagnosed theory-practice gap (Syed et al., 2018) by consolidating the highly dispersed body of research on the challenges to KM implementation to provide an evolutionary overview on the most persistent challenges, their forms of appearance, proposed solutions, and their historical roots. In this section, we present and discuss a genealogical model that traces the historical development paths of the fundamental root challenges and theorise generic principles according to which the scholarly discussion developed.

5.1. Historical development path model of KM challenge evolution

Figure 7 presents our consolidation of the traces of KM implementation challenges identified from the literature review into a historical development paths model across all three decades (leaving out the last 4 years as they only give us a trend but do not represent a decade). It also acts as a novelty matrix according to Castaneda and Cuellar (2020) by indicating whether the KM challenge topics are emerging, established, or declining.

We witness that the discussion of KM challenges at the beginning of the first decade in the early to the mid-1990s remained at a generic and strategic level.

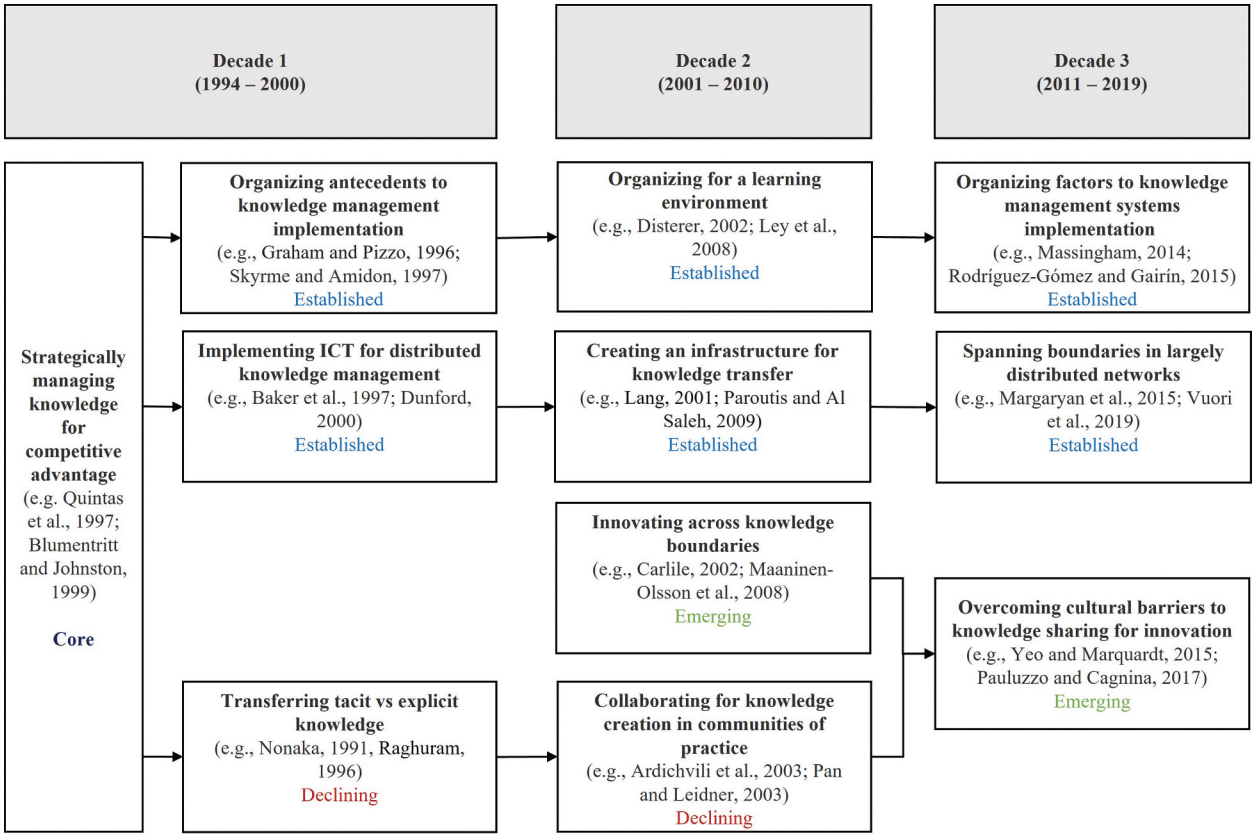


Figure 7. Historical development path Model of KM challenges 1991–2019.

The explosion of the publications on a topic that was beforehand quasi non-existent (Wilson, 2002) and the issued strategic-level claims about knowledge being a firm's most valuable resource for achieving competitive advantage (e.g., Jordan & Jones, 1997; Nonaka, 1991) that mark the beginning of this discourse made it resembles a management fad fulfilling "symbolic functions such as signalling innovativeness" (Abrahamson, 1991, p. 588). The topic, however, entered the management discourse from an important societal debate that started in 1973 when Daniel Bell announced "the coming of the post-industrial [or knowledge, the authors] society" (Grusky, 2001) and argued that knowledge would play an increasingly important role in this future (Hislop et al., 2018).

Our study identifies that once the predicted growth of service-related knowledge work exceeded the expectations in the late 1990s (Castells, 1998; Kumar, 1995), firms took the topic up as a strategic management theme, while academics anchored it on an emergent new theoretical perspective, the resource-based view (Barney, 1991). This substantiated the field: Our model shows that instead of ending as a management fad, the strategic discourse then unfolded into three more concrete KM implementation challenges: 1) Organisational antecedents to implement a knowledge-sharing culture (Graham & Pizzo, 1996; Skyrme & Amidon, 1997), 2) the use of ICT for knowledge transfer (Baker et al., 1997; Goodman & Darr, 1998), and 3) knowledge conversion (Blumentritt & Johnston, 1999; Nonaka, 1991; Raghuram, 1996). What our model demonstrates is that these three topics guide the evolution of KM challenge implementation topics across all three decades, thereby providing the field with focus, but also creating historical path-dependencies in the discussion. All three are of importance for the evolution of the field, and in demonstrating this, our findings contradict those by Shujahat et al. (2019, p. 442) who have identified an "overarching role of the human and cultural approach to knowledge management over the Big Data and IT and system-based approaches".

The novelty matrix Castaneda and Cuellar (2020) yet also allows us to highlight that the KM implementation challenges that relate to organisational antecedents and ICT remain established topic areas in the discourse throughout three decades that both increase their focus and breadth and thereby created new challenges that could be handled with adapted solutions. Instead, the topic of knowledge conversion renewed itself by merging with innovation-related topics (e.g., Carlile, 2002; Yeo & Marquardt, 2015) and thus required partly also new solutions. The preliminary analysis of the first four years of the fourth decade indicates that ICT continues to be a significant topic in knowledge management. However, there is a noticeable shift in context; there is now an increased focus on managing knowledge risks (e.g., associated with shadow ICT: Abbas & Alghail, 2023), as well as on the growing role of artificial intelligence as an emerging technology to support knowledge management processes, such as knowledge sharing, creation and transfer (Ali et al., 2023; Nylund et al., 2023). The implementation challenges and organisational antecedents to KM no longer constitute a separate cluster; instead, they are implicitly represented across all clusters. For instance, this can be seen in the context of ensuring the safe and responsible use of ICT (Abbas & Alghail, 2023, Cluster 1) or in the adoption of AI-assisted knowledge management practices (Moldt et al., 2023; Nylund et al., 2023, Cluster 2). The emphasis on knowledge conversion within the realms of innovation and culture has decreased. In relation to this thematic cluster, a shift is noticable towards challenges associated with social media and technology in knowledge sharing (Igwe & Olatokun, 2023; Zhang et al., 2022) and in the stakeholder engagement for knowledge creation for sustainable development (Patra et al., 2021; Venegas et al., 2021).

Our genealogic perspective thus goes beyond simply summarising the discussion in the field. It instead allows us to highlight where research aimed at advising practice can adapt established solutions that were earlier used to solve the same root challenge with a slightly different focus, and where finding a solution to an emerging KM challenge requires further research because the "root 'tributaries'" (Rasche & Chia, 2009, p. 714) of a discourse change.

In this, we extend earlier research that studied the evolution of KM as a research field (Gaviria-Marin et al., 2019) not only by focussing on KM challenges and solutions, but also by identifying areas where future research would be most needed to create new solutions because two root challenge topics merge or new ones emerge. We contribute to directing solution development by clearly indicating the origin of current KM challenges based on a genealogical study. Further, this evolutionary perspective allows us to not just classify KM challenges at a given point in time according to content-related criteria such as people-related, technology and organisational challenges (e.g., Durst & Zieba, 2019; Hislop et al., 2018), implementation process phase of appearance (e.g., knowledge creation, sharing, transferring, or retention) or organisation type (e.g., Durst & Bruns, 2018; Soakell-Ho & Myers, 2011). Instead, we additionally bring in the historical perspective on form and roots of KM challenges, as called for by Maclean et al. (2016) for historic organisational studies.

5.2. Generic principles of the evolution of the scholarly debate

We find that the three topics reappear across the decades and, in general, become more concrete: For example, the stream on organising antecedents to KM implementation creates general management frameworks during the first decade, moves on to the challenges of organising learning environments in the organisation in the second decade and finally studies challenges related to the implementation of specific KM systems in the third decade. In addition, our model points towards a shift from a narrow to a broad challenge and solution focus. An example is the shift from focusing solely on ICT and infrastructure to knowledge transfer (decade 2) towards addressing the bigger challenge in which ICT can be used as one tool to span boundaries in large distributed networks in decade 3 (Haas & Cummings, 2015; Hwang et al., 2015; Margaryan et al., 2015). These insights extend those of existing literature review-based studies that normally go not beyond proving the existence and describing topical generations in the Knowledge Management field (Gaviria-Marin et al., 2019) by demonstrating, for the topical area of KM implementation challenges, consistency of topical development across such generations through evincing clear topical development trajectories. For the specific literature stream on KM challenges, we uncover the roots of the in the discourse used (common sense) categorisation of systems-, -technology- and people-related challenges (Durst & Zieba, 2019; Hislop et al., 2018; Syed et al., 2018). Yet, our review shows that there are, from the second decade on, crossovers between the challenge categories – the body of literature on the implementation of KM systems in the third decade, for example, includes challenges within the system, the technology, and people categories (Pandey & Dutta, 2013).

Our data allow for reconstructing the emergence of two new challenges: Innovating across knowledge boundaries (decade 2) and retaining knowledge in organisations (decade 3). Both have their origins at the point in time of their emergence hyped societal discourse: the shift towards a globalised environment (decade 2) and the ageing society (decade 3). The discourse about globalisation gained momentum with China's enrolment in the World Trade Organization (WTO) in December 2001 (WTO, 2024) as a transformational moment in shifting the economy towards a global economy. We witness that specifically around 2001, the challenges that the shift towards globalisation poses for KM entered the research stage (Davis et al., 2005; Garud & Kumaraswamy, 2005; Lang, 2001; Majchrzak et al., 2005), with the number of publications sharply increasing after 2004. These specifically relate to knowledge creation and knowledge transfer in globalised or dispersed teams (e.g., Carlile, 2002; Maaninen-Olsson et al., 2008). We see the same path with the topic of knowledge retention in the third decade: The trending discussions that started in the last years of the 2000er decade about the trend towards an ageing (Western) society where the number of older people will increase while the number of young professionals will decrease draw the attention of KM practitioners and academics on the need to retain the knowledge of the retiring workforce (Levy, 2011; Ragab & Arisha, 2013).

Following this trend, the number of publications on knowledge retention in Scopus shows a significant increase after 2010. Together with the insights on the emergence of the KM challenge topic in the first decade, we extend recent insights from the literature review with the finding that hyped societal discourses can act as triggers for a shift of the attention focus in management practice and research that enlarges the scope of the respective topic and constitutes a new arena for discourses about challenges and solutions. We yet also recognise that the new challenge in the second decade – innovating across boundaries – was integrated with the existing perspectives through a merger with another cluster – collaborating for knowledge creation in CoPs – whose challenges were extensively studied and detailed solutions were available by the end of that decade (e.g., Wolf et al., 2011), but we cannot conclude from our sample whether this is a general pattern of embedding new topics.

6. Conclusion

The study presented in this paper advances research by responding to the calls for a consolidation of the highly dispersed body of research on the challenges to KM implementation (e.g., Durst & Runar Edvardsson, 2012; Lee & Chen, 2012). It provides an evolutionary overview on the most persistent KM challenges, proposed solutions, and their historical roots and forms of appearance. The study offers a decade-wise overview of the highly dispersed literature on KM implementation challenges published between 1991 and 2023 and uses a genealogical model to identify and visualise 1) Organisational antecedents to implement a knowledge-sharing culture (Graham & Pizzo, 1996; Skyrme & Amidon, 1997), 2) the use of ICT for knowledge transfer (Baker et al., 1997; Goodman & Darr, 1998), and 3) knowledge conversion (Blumentritt & Johnston, 1999; Nonaka, 1991; Raghuram, 1996) as the three topics guide the evolution of KM challenge implementation topics across all three decades.

In going beyond a simple description and listing of KM challenges and solutions, it identifies historical emergence patterns of the most debated KM challenges. Based on the resulting evolutionary model, it

theorises generic genealogic principles according to which the debate about KM implementation challenges developed so far, such as the shift from a narrow to a broad challenge and solution focus. The study contributes to bridging the theoretical practice gap by visualising the most persistent KM implementation challenge streams and solutions. Our model can be used by academics in boundary spanning activities to legitimise (Ungureanu & Bertolotti, 2020) and mobilise (Alsamarra'i, 2019) knowledge movements through interactions between academia and practice.

6.1. Limitations and future research avenues

As with most research, this study has its limitations. First, we included only articles from the Scopus database into this review. Although the number of papers obtainable from the Scopus database correlates to "R2 ≈.99" (Archambault et al., 2009, p. 1320) with those available from the Web of Science, it is still possible that we missed out on articles published in other databases like EBSCO, Web of Science, or Google Scholar. We tried to mitigate this limitation by ensuring that our data is up to date. Further, we included only articles until 2019, which makes sense for our decade-wise perspective, but might distract our attention from KM challenges that cross decades and become only blurredly visible. With emerging technologies such as AI changing the topical area of KM greatly within only a few years, we recommend follow-up studies with different analysis time frames.

Furthermore, our research identifies potential avenues for future research that we believe could be particularly helpful for closing the theory-practice gap and beneficial for practitioners as findings might inform approaches for solving either recent or upcoming KM challenges. A recent challenge that practitioners face focuses on retaining knowledge. The importance of this challenge and the need for research on it is amplified by a shift in the labour market towards an increasing number of employees voluntarily quitting their jobs, reflecting "the great resignation" (Serenko, 2022). We thus second recent calls for research to identify and test knowledge retention strategies with a volatile workforce (Hirsch, 2021; Serenko, 2022; Sheather & Slattery, 2021; Sull et al., 2022).

An upcoming KM challenge emerges from the knowledge risks associated with digitalisation (Durst & Zieba, 2019). While so far, KM was developing theory and practical approaches to identify the most effective methods for leveraging the organisation's knowledge base (Durst & Runar Edvardsson, 2012; Massingham, 2014; Stam, 2009), "knowledge risk is a measure of the probability and severity of adverse effects of any activities engaging or related somehow to knowledge that can affect the functioning of an organization on any level" (Durst & Zieba, 2019, p. 2). The dark side of KM and digitalisation is an area of research that has so far focused on the risks of data theft and ignored the social impact of digitalisation, for example on knowledge sharing. We call here for conceptual contributions that theorise different scenarios of KM in a digitalised organisation where humans are mainly involved in decision and maintenance tasks (Agrawal et al., 2019).

Our research identifies the persistence of KM challenges related to the establishment of organisational antecedents for KM implementation, ICT infrastructure and use, and knowledge conversion as equally important to the evolution of the field. In this, we contradict Shujahat et al. (2019) who identify human and cultural approaches as more important than IT system-based approaches to ensure efficient KM implementation. We call for future studies to investigate these contradicting findings.

Further, our analysis of the literature from the first half of the recent decade indicates three main emerging topics as key research areas: In the information technology stream, the potentials, challenges, and enabling conditions for using artificial intelligence (e.g., natural language processing through language models) in knowledge management processes must be conceptualised and evaluated. In the knowledge conversion stream, the creation of knowledge for sustainability, specifically the co-creation of knowledge with an emphasis on stakeholder engagement is an upcoming topic that researchers are highly recommended to focus on. Combining these two topics, future research should aim to understand and conceptualise the challenges in human collaboration with artificial intelligence. This could include areas such as knowledge sharing and collaborative knowledge creationfor example, in the context of innovation or addressing sustainability challenges.

6.2. Implications for improving research-firm interaction and closing the theory-practice GAP

Our findings suggest that in the first phase of the emergence of a KM challenge cluster, a hyped issue of societal concern triggers the emergence of a field of interest to management scholars and practitioners. During this inception phase, researchers and practitioners mutually validate the importance of the topic for theory development and competitive advantage, and thereby legitimising further engagement and initiatives. At the end of this phase, management research is capable of providing practitioners with first generic frameworks for their strategic initiatives.

The theory-practice gap (Syed et al., 2018) opens in the second phase when managers develop solutions to the challenges in trial-and-error approaches, while

researchers mainly study and observe the phenomenon. Notwithstanding the usefulness of research and the theoretical framing that happens, the research findings that are generated are often only an ex-post explanation to the practitioners as to why their initiatives have been successful. This is precisely the point of the complaints that research on KM challenges is too conceptual and does not have enough concrete evidence to guide practice (Booker et al., 2008; Ragab & Arisha, 2013; Serenko, 2021; Verma et al., 2020). Instead, in the third phase, when there is a need to transfer the solutions developed to other contexts or to modified or broader challenges, researchers can guide the efforts of practitioners by studying the application of solutions under modified conditions.

What we are proposing here, then, is greater collaboration between academics and researchers in the second phase of the development of a KM challenge (cluster) theme. This would argue in favour of the field studies proposed by Serenko (2013), if they were designed for substantial collaboration, communication, and irritation between researchers and managers immediately after the basic strategic frameworks have been developed, and the agenda set in the companies.

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Data availability statement

To support the findings of the qualitative content analysis, the bibliographic information of all included articles per decade and decade cluster have been compiled in a file, which is openly available on Mendeley Data at doi:10.17632/hxywx67bz7.1.

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